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Interview-Specific Insights and Focus Areas

In Progress

AIML Interview Preparation



Reading Material

⌚ 2 min read

Frequently tested patterns:

1. **Reshape trap:** Interviewers ask candidates to reshape an array and check if they verify element count consistency.
2. **Axis confusion:** Asking `np.mean(arr, axis=0)` vs `axis=1` on a 2D array. Always verify: axis=0 collapses rows, axis=1 collapses columns.
3. **@ vs * operator:** Candidates who confuse element-wise and matrix multiplication are flagged immediately in DS/analytics roles.
4. **Broadcasting in ML context:** "How does gradient descent update weights across a batch?" — the answer involves broadcasting salary vectors and weight matrices.
5. **argmax/argsort:** Often used in recommendation ranking and top-k selection.
6. **keepdims trap:** Candidates who reshape manually after an aggregation instead of using `keepdims=True` are marked down for missing the idiomatic broadcasting-friendly approach.
7. **np.where vs. boolean indexing:** Mixing up "extract matching elements" (boolean indexing) with "replace non-matching elements, same shape" (`np.where`) is a common conceptual slip probed in interviews.

Common interview question themes:

- "Write NumPy code to normalize a feature column to [0,1] range."
- "Given two matrices, compute cosine similarity using NumPy."
- "Why would you use `np.vectorize()` vs a list comprehension?"

- "Given an array of labels, find which class has the fewest samples using NumPy only."

Red flags interviewers look for:

- Looping through array elements when vectorized operations exist.
- Ignoring dtype when doing divisions (integer array divided by integer = integer result in older NumPy versions).
- Not checking shape before matrix multiplication.

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